Rita Bhowmik

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Education

University Of Waterloo Waterloo, ON

Mathematics & Economics *Honours, Co-op - 2A Term* *Sep. 2025 – May 2030*

- **Coursework in** Calc I & II, Computing, Physics, Microeconomics, Financial Analysis
- **Community Involvement** Waterloo Tech Week, Research @ C&O Department

Professional Experience

Women In Math - Undergraduate Research Assistant Sep 2026 – Present

University Of Waterloo - Maths Department *Waterloo, ON*

- Researching social networks through c-closed graphs, looking at finding subgraphs that represent community structures, among other classical graph theory problems. Investigating how the difficulty of vertex coloring is affected by the c-closed property.
- Collaborated with PHD students in Combinatorics & Optimization department.

Engineering Intern May 2026 – August 2026

Mespere Lifesciences *Waterloo, ON*

- Worked within a technical, fast-paced team designing real-world sensors. Supported manufacturing engineering of R1 and R2 sensor lines, analyzing data and processes to improve reliability and scalability.
- Collaborated with cross-functional teams (software, business, and data) to research sensors with NeuroOS and V0100 software platforms.
- Conducted R&D on photonic hemodynamic sensors, evaluating performance characteristics and contributing to mathematical design improvements for medical device systems

Marketing Analyst June. 2024 – Sep. 2025

Complete Eyecare Optometry *Mississauga, ON*

- Analyzed historical marketing and patient engagement data to design and execute a strategy that generated 40K+ new impressions and increased outreach by 900% over 8 months.
- Implemented SEO optimization strategies, improving search visibility and driving measurable increases in inbound traffic.
- Built and managed analytics dashboards across 4 platforms to track campaign performance and inform data-driven decisions.

Awards & Projects

1st Place - Jane Street Estimathon @ Hack the 6ix | *Fermi estimation, Probability* July 2026

- Won 1st place in the Jane Street Estimathon at Toronto's Hack the 6ix, demonstrating strength in probability, mathematical modeling, and data-driven estimation under strict time constraints.
- Placed 1st out of 120+ competitors, applying probabilistic reasoning, estimation techniques, and quantitative problem-solving.

IMC Prosperity Python Algorithm | *Python, VSCode, Git, Matplotlib, Numpy* Nov 2025

- Developed algorithmic trading strategies in Python using NumPy and quantitative modeling techniques to optimize decision-making under uncertainty.
- Simulated market conditions and evaluated strategy performance using data driven back testing and visualization.

Skills

Analytics & Data: Python (Pandas, NumPy, Matplotlib), SQL, R, Excel, AWS AI Practitioner (certified)
Quantitative: Statistics, Economics, Financial Modeling, Loan Amortization
Business & Ops: Stakeholder Communication, Project Ownership, Process Design, Growth and Logistics, Consulting